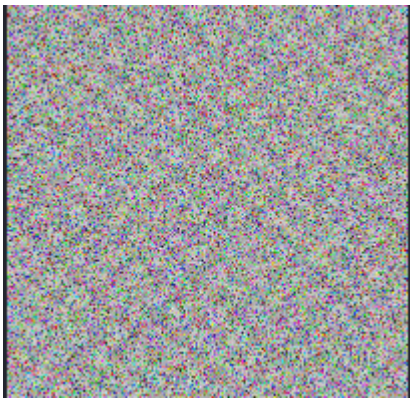
crossed-pil misc challenge UNR

We receive an image:



First though it LSB. Ran zsteg and didn’t find much.
Then performed strings, and right at the end saw this script which is probably how this image was obtained:

```
import numpy as np
from PIL import Image
import random
img = Image.open('flag.png')
pixels = list(img.getdata())
oioi=[]
for value in pixels:
    oi = []
    for oioioi in value:
        # hate me note for the var names ;)
        if oioioi == 255:
            oioioi = random.choice(range(0, 255, 2))
        else:
            oioioi = random.choice(range(0, 255, 1))
        oi.append(oioioi)
    oioi.append(oi)
img = Image.new('RGBA', [200,200], 255)
data = img.load()
count = 0
for x in range(img.size[0]):
    for y in range(img.size[1]):
        data[x,y] = (
            oioi[count][0],
            oioi[count][1],
            oioi[count][2],
            oioi[count][3],
        )
        count = count + 1

img.save('image.png')
```

Now, to be honest I found it kinda difficult to understand it, since I didn't work with numpy and pillow that much. So I got help from Claude to write me a reverse script:

```
from PIL import Image

img = Image.open('image.png')
pixels = list(img.getdata())
new_pixels = []

for pixel in pixels:
   new_pixel = []
   for channel in pixel:
       if channel % 2 == 0:
           new_pixel.append(255)  # was white
       else:
           new_pixel.append(0)    # was not white
   new_pixels.append(tuple(new_pixel))

out = Image.new('RGBA', img.size)
out.putdata(new_pixels)
out.save('flag_recovered.png')
```

And we got this:

I tried doing different things to blacken out those non white pixels, but it wasn't enough for a qr code reader to detect it. So I did that manually, using GIMP. Scanned the QR code and got the flag.

flag:ctf{3c7f44ab3f90a097124ecedab70d764348cba286a96ef2eb5456bee7897cc685}

Note: there was an alternative solution to strings and reversing the script. stegsolve performs LSB and offers a preview. And for a specific one, you could see that there is a QR code

hidden.:

And from here, pretty much the same, manually “painting” the QR code.